

Last Time

Random Variables

Distributions

Discrete Distributions

Continuous Distributions

Relationships Between Distributions

QQ Plots

Today

Distributions

- Discrete Distributions

- Continuous Distributions

Relationships Between Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Sampling Distribution of $\bar{X}$

Distributions

Discrete	Continuous
Uniform	Uniform
Bernoulli	Normal
Binomial	Student's t
Poisson	χ^2
Tabled	F

Uniform

Roll a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$m = |S| = 6$$

$$p_X(x|m) = \frac{1}{m} \text{ for } x \in S, 0 \text{ else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b-a+1} \text{ for } x \in S, \text{ 0 else}$$

$$E(X|a, b) = \frac{a+b}{2}$$

$$\text{Var}(X|a, b) = \frac{(b-a+1)^2 - 1}{12}$$

Bernoulli

A single experiment with only two possible outcomes: success and failure.
Flip a coin, call “heads” a success.

$$\begin{aligned} S &= \{0, 1\} \\ 0 &\leq p \leq 1 \\ p_X(x|p) &= \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned} E(X) &= p \\ \text{Var}(X) &= p(1 - p) \end{aligned}$$

Binomial

Add up a sequence of N independent Bernoulli (0 or 1) experiments: flip a coin N times, answer N true/false questions. How many “successes” were there in the sequence?

$$S = \{0, 1, \dots, N\}$$

$$0 \leq p \leq 1$$

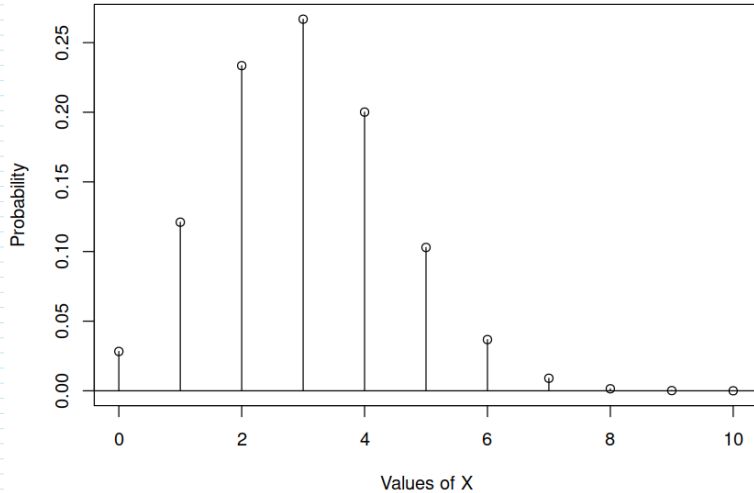
$$N \geq 1, N \in \mathcal{N}^+$$

$$p_X(x|N, p) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x} & x \in S \\ 0 & \text{otherwise.} \end{cases}$$

$$E(X) = Np$$

$$\text{Var}(X) = Np(1-p)$$

Binomial



Poisson

- ▶ The number of rare events occurring in a fixed interval of time or bounded region of space. Airline accidents, hospital admissions, school shootings, typos on a web page, defective widgets in a shipment, etc.
- ▶ The limiting distribution of the binomial when $N \rightarrow \infty$ and $p \rightarrow 0$.

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$$S = \{0, 1, 2, \dots\}$$

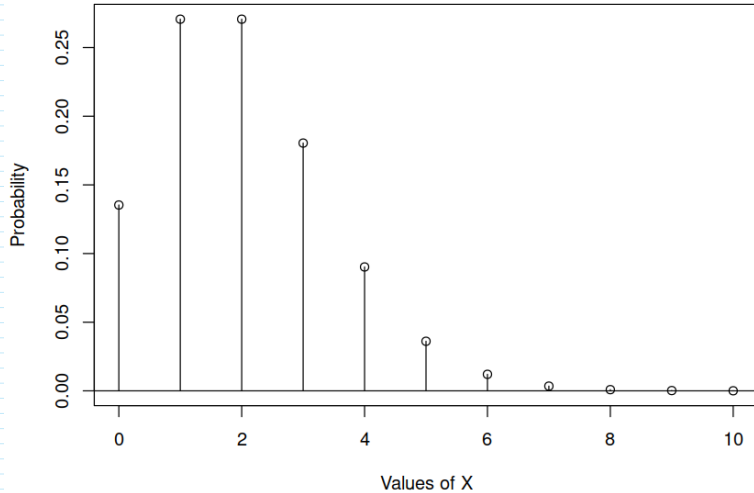
$$\lambda > 0$$

$$p_X(x|\lambda) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x \in S \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

Poisson



Tabled

A random variable that can take on values in a list, with each value x associated with a probability π_x . Relative frequency distributions are examples of tabled probability distributions.

GSS Health Rating Responses

x	1	2	3	4
π_x	0.37	0.46	0.15	0.02

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x	1	2	3	4
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$$S = \{1, 2, 3, 4\}$$

$$p_X(x|\pi_1, \pi_2, \pi_3, \pi_4) = \begin{cases} \pi_1 & x = 1 \\ \vdots & \\ \pi_4 & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \sum_{x=1}^N x\pi_x$$

$$Var(X) = \sum_{x=1}^N (x - E(X))^2 \pi_x$$

Uniform

- ▶ A “rectangular” distribution with support equal to a range; all values in the range have equal density.
- ▶ Important for statistical applications: generation of random numbers, Bayesian priors

$$\begin{aligned} S &= [a, b] \\ a &< b \\ f_X(x|a, b) &= \begin{cases} \frac{1}{b-a} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= \frac{a+b}{2} \\ \text{Var}(X) &= \frac{(b-a)^2}{12} \end{aligned}$$

Normal

- ▶ The most important distribution for us!
- ▶ Many natural systems result in normally distributed variables.
- ▶ Central Limit Theorem

$$S = (-\infty, \infty) = \mathcal{R}$$

$$\mu \in \mathcal{R}$$

$$\sigma^2 > 0$$

$$f_X(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for } x \in S$$

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$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

Standard Normal

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\text{Z-score of } X: Z_X = \frac{X - \mu}{\sigma}$$

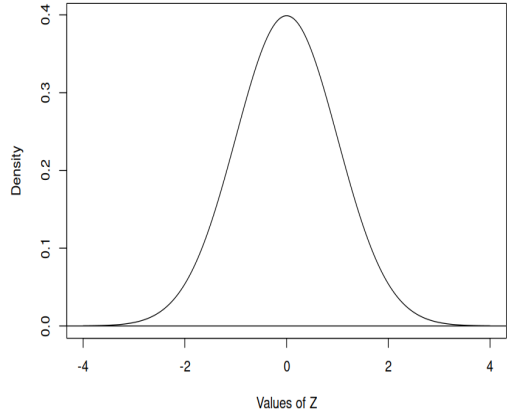
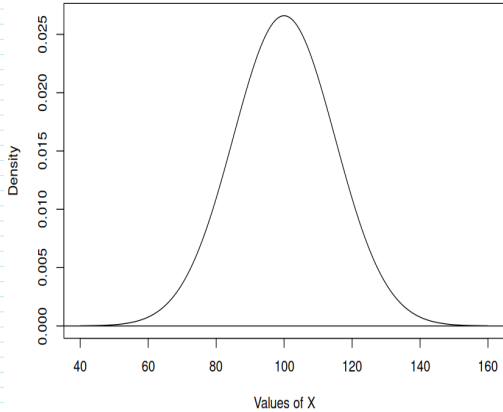
$$S = (-\infty, \infty) = \mathcal{R}$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \text{ for } z \in S$$

$$E(Z) = 0$$

$$\text{Var}(Z) = 1$$

Normal and Standard Normal



Student's t

- ▶ William Sealy Gosset (1876-1937)
- ▶ A bell-shaped distribution used when the population variance σ^2 is estimated by the statistic s^2 .

$$\begin{aligned} S &= (-\infty, \infty) = \mathcal{R} \\ \nu &\in \{0, 1, 2, \dots\} \\ f_X(x|\nu) &= \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad \text{for } x \in S \end{aligned}$$

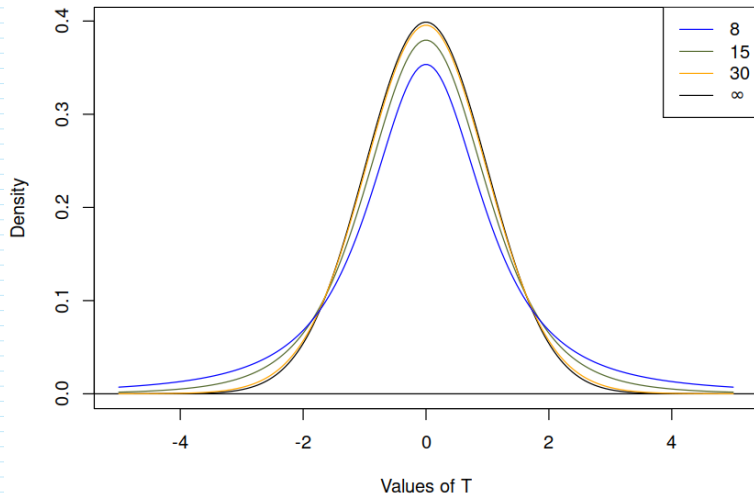
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$$\begin{aligned} E(X) &= \begin{cases} 0 & \nu > 1 \\ \text{undefined} & \text{otherwise} \end{cases} \\ \text{Var}(X) &= \begin{cases} \frac{\nu}{\nu-2} & \nu > 2 \\ \infty & 1 < \nu \leq 2 \\ \text{undefined} & \text{otherwise} \end{cases} \end{aligned}$$

Student's t



χ^2

The distribution of a sum of squared z -scores, related to the distribution of the sample variance.

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ k &\in \{1, 2, \dots\} \\ f_X(x|k) &= \begin{cases} \frac{x^{k/2-1}}{2^{k/2}\Gamma(k/2)} e^{-x/2} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

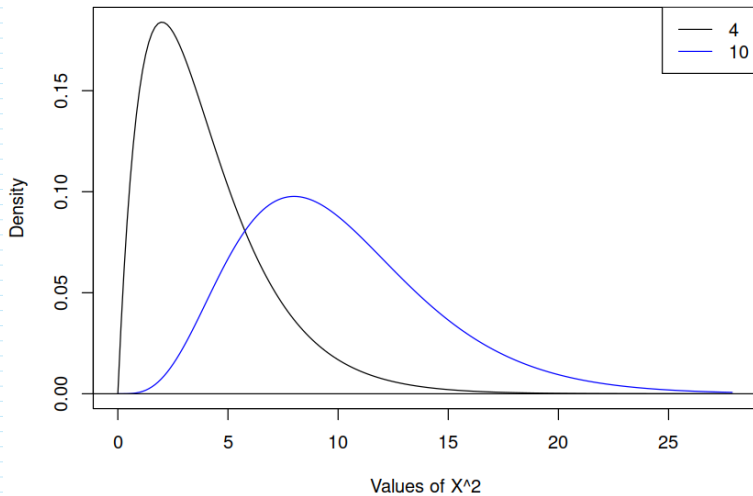
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$$\begin{aligned} E(X) &= k \\ \text{Var}(X) &= 2k \end{aligned}$$

χ^2



F

- ▶ A distribution related to the ratio of two sample variances,
- ▶ The basis for ANOVA, regression model comparison, etc

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ \nu, \delta &> 0 \\ f_X(x|\nu, \delta) &= \begin{cases} \frac{\Gamma\left(\frac{\nu + \delta}{2}\right) \nu^{\nu/2} \delta^{\delta/2}}{\Gamma\left(\frac{\nu}{2}\right) \Gamma\left(\frac{\delta}{2}\right)} \frac{x^{\nu/2-1}}{(\delta + \nu x)^{(\nu+\delta)/2}} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

F

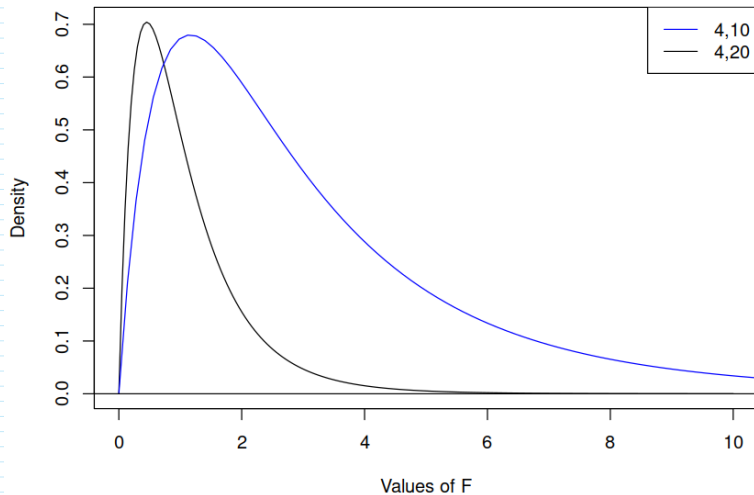
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$$\begin{aligned} E(X) &= \frac{\delta}{\delta - 2} \text{ for } \delta > 2 \\ \text{Var}(X) &= \frac{2\delta^2(\nu + \delta - 2)}{\nu(\delta - 2)^2(\delta - 4)} \text{ for } \delta > 4 \end{aligned}$$

If $\delta \leq 2$ the mean does not exist. If $\delta \leq 4$ the variance doesn't exist.

F



Notation

Discrete	Continuous
Uniform $U(a, b)$	Uniform $U(a, b)$
Bernoulli $Ber(p)$	Normal $\mathcal{N}(\mu, \sigma^2)$
Binomial $Bin(N, p)$	t $t(\nu)$
Poisson $Pois(\lambda)$	χ^2 $\chi^2(k)$
Tabulated $p_X(x \pi_1, \dots, \pi_m)$	F $F(\nu, \delta)$

Relationships Among Distributions

Leemis Chart

- ▶ $X \sim \text{Ber}(p): \sum_{i=1}^N X_i \sim \text{Bin}(N, p)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty, Np = \lambda} \text{Bin}(N, p) \sim \text{Pois}(\lambda)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty} \text{Bin}(N, p) \sim \mathcal{N}(Np, Np(1 - p))$
- ▶ $Z \sim \mathcal{N}(0, 1): \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$
- ▶ $X \sim t(\nu): X^2 \sim F(1, \nu)$
- ▶ $X \sim t(\nu): \lim_{\nu \rightarrow \infty} t(\nu) \sim \mathcal{N}(0, 1)$
- ▶ $X^2 \sim \chi^2(k): \lim_{k \rightarrow \infty} \chi^2(k) \sim \mathcal{N}(k, 2k)$
- ▶ $X_1^2 \sim \chi^2(n_1), X_2^2 \sim \chi^2(n_2): \frac{X_1^2/n_1}{X_2^2/n_2} \sim F(n_1, n_2)$

QQ Plot

- ▶ Plot the quantiles of the observed data (y axis) against the quantiles of the theoretical distribution (x axis).
- ▶ The plot points should fall on the identity line $y = x$.

R code

```
data <- read.csv("GSS_1980.csv",header=TRUE)
age <- data$AGE

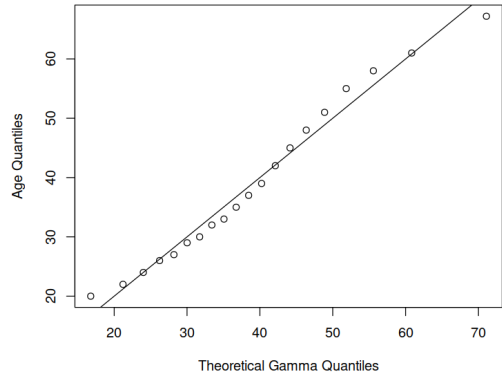
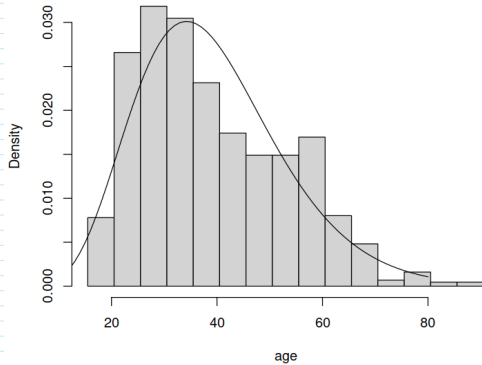
# Pick some good values for the gamma distribution
a <- mean(age)/var(age)
k <- mean(age)^2/var(age)

# Compute the quantiles for the data and the gamma model
obs <- quantile(age,seq(.025,.975,by=.05))
theory <- qgamma(seq(.025,.975,by=.05),k,a)

# Plot the data histogram with the gamma model
par(mfrow=c(1,2))
hist(age,freq=FALSE,breaks=seq(15.5,90.5,by=5),
      xlab="age",ylab="Density",main="")
lines(0:80,dgamma(0:80,k,a))

# QQ Plot
plot(theory,obs,type="p",ylab="Age Quantiles",
      xlab="Theoretical Gamma Quantiles")
abline(0,1)
```


QQ Plot



Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

Family	Sibling	
	Older	Younger
1	14	32
2	59	85
3	89	98
4	88	08
5	47	20
6	98	94
7	36	78
8	84	14
9	55	38
10	10	81
11	26	20
12	38	27
13	51	35
14	81	32
15	99	76

Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
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11	26	20	+
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13	51	35	+
14	81	32	+
15	99	76	+

Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?
- ▶ $Pr(X \geq 10 | p = .5) = 0.15$

```
N <- 15  
p <- 0.5  
1 - pbinom(9, N, p)
```

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
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9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Questions/Goals

- ▶ Scientific Question: Are older siblings more conservative than younger siblings?
- ▶ Statistical Question: Is the number of “+”s we observed likely to have occurred by chance?

Sampling Distributions

```
# Draw samples from an Exp(1) population
plot(seq(0,5,length=200),dexp(seq(0,5,length=200)),
     type='l',xlab = "Values of X",ylab="Density")

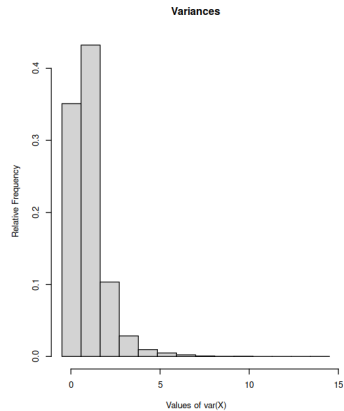
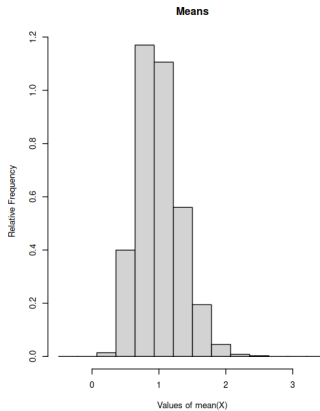
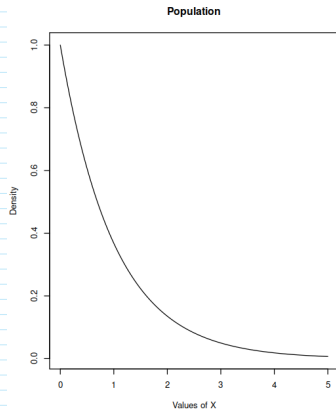
bignum <- 10000
N <- 10
means <- vector()
vars <- vector()

for (i in 1:bignum) {
  X <- rexp(N)
  means[i] <- mean(X)
  vars[i] <- var(X)
}

par(mfrow=c(1,2))
hist(means,breaks=seq(-.5,round(max(means))+.5,length=15),freq=FALSE,
     ylab="Relative Frequency",xlab="Values of mean(X)",main="Means")

hist(vars,breaks=seq(-.5,round(max(vars))+.5,length=15),freq=FALSE,
     ylab="Relative Frequency",xlab="Values of var(X)",
     main="Variances")
```

Sampling Distributions



Sampling Distribution

Definition (Sampling Distribution)

A *sampling distribution* is the distribution of possible values of $S(X)$ under all possible random samples of X of a fixed size N .

Distributions of Statistics

Single samples of size N

Parameter	Test Statistic	Distribution
μ	$\bar{X}$	$\mathcal{N}(\mu, \sigma^2/N)$
μ	$\frac{\bar{X} - \mu}{s/\sqrt{N}}$	$t(N-1)$
π	p	$\mathcal{N}(\pi, \pi(1-\pi)/N)$
σ^2	$\frac{(N-1)s^2}{\sigma^2}$	$\chi^2(N-1)$
ρ	$\frac{1}{2} \ln[(1+r)/(1-r)]$	$\mathcal{N}(\tanh^{-1}(\rho), 1/\sqrt{N-3})$

The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.

As N gets large (goes to infinity), the shape of the distribution of the sample mean $\bar{X}$ will become normal.

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Formally: For an iid sample $X = \{X_1, X_2, \dots, X_N\}$ of size N drawn from a population with mean μ and standard deviation σ , define the random variable $S_N = \sum_{i=1}^N X_i$.

$$\lim_{N \rightarrow \infty} Pr\left(\frac{S_N - N\mu}{\sigma\sqrt{N}} \leq x\right) = Pr(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz.$$

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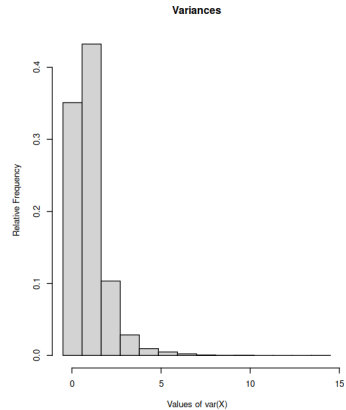
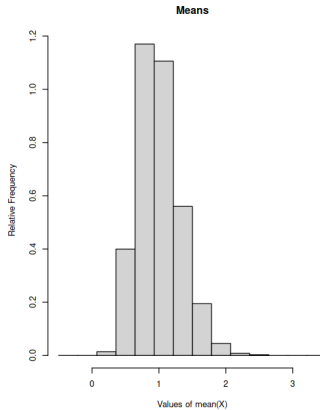
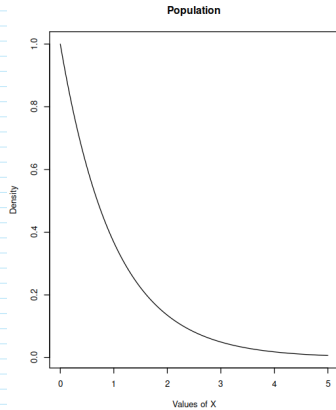
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$$\lim_{N \rightarrow \infty} Pr\left(\frac{S_N - N\mu}{\sigma\sqrt{N}} \leq x\right) = Pr(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz.$$

Regardless of the shape of the distribution of X !

Exponential Population



Central Limit Theorem

Why is the CLT important?

Central Limit Theorem

Why is the CLT important?

- ▶ It is the foundation of statistical inference across many different disciplines, data types, and analytic techniques.
- ▶ Many statistical procedures rely on assumptions of normality, which the CLT guarantees.
- ▶ We know how $\bar{X}$ will be distributed even if we know nothing about the population from which X was sampled.
- ▶ We know that the mean of $\bar{X}$ is equal to the population mean μ , which is the thing we want to tell stories about.

Normal Random Variables

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

- ▶ $X + c \sim \mathcal{N}(\mu + c, \sigma^2)$
- ▶ $cX \sim \mathcal{N}(c\mu, c^2\sigma^2)$

If X_1 and X_2 are independent and $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, then

- ▶ $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
- ▶ $aX_1 + bX_2 \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$

If $X = \{X_1, X_2, \dots, X_N\}$ is an iid sample with $X_i \sim \mathcal{N}(\mu, \sigma^2)$ then

- ▶ $\sum_{i=1}^N X_i \sim \mathcal{N}(N\mu, N\sigma^2)$

If some number of independent normal random variables are linearly transformed or combined, the result is also a normal random variable.

The Sampling Distribution of the Mean

By the CLT, we know $\bar{X} \sim \mathcal{N}(\mu_{\bar{X}}, \sigma_{\bar{X}}^2)$

What are $\mu_{\bar{X}}$ and $\sigma_{\bar{X}}^2$?

$\mu_{\bar{X}}$

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ &= \frac{1}{N} E\left(\sum_{i=1}^N X_i\right) \\ &= \frac{1}{N} \left(\sum_{i=1}^N E(X_i)\right) \\ &= \frac{1}{N} \left(\sum_{i=1}^N \mu\right) \\ &= \frac{1}{N} N\mu = \mu. \end{aligned}$$

$$\sigma_X^2$$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ &= \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N X_i\right) \\ &= \frac{1}{N^2} \left(\sum_{i=1}^N \text{Var}(X_i)\right) \\ &= \frac{1}{N^2} \left(\sum_{i=1}^N \sigma^2\right) \\ &= \frac{1}{N^2} N \sigma^2 = \frac{\sigma^2}{N}. \end{aligned}$$